

Expanded Mathematical Appendix

Scalable Spatial Model for Brain MRI Lesion Mask Data

Companion to `cbmr_paper.pdf`

15 September 2026

Abstract

This self-contained appendix develops the model in the supplied manuscript by Yifan Yu and Thomas E. Nichols, *Scalable Spatial Model for Brain MRI Lesion Mask Data*. Its central parameters are spatially varying covariate-effect maps for binary subject images. We derive the Poisson and independent negative-binomial working scores and Hessians, the separable initialization, the Kronecker preconditioner, the subject-centered Taylor approximation, and model-based and subject-clustered inference. The supplement’s QR sandwich and toy slope-map variance are expanded, including their coefficient-order and dimension qualifications. The manuscript’s mass-univariate logistic comparator is treated separately. All intermediate dimensions and vectorization conventions are explicit. Where a displayed manuscript formula needs qualification or correction, that distinction is stated rather than inherited silently. This is a handwritten mathematical review document, not a new claim of complete symbolic or proof-assistant certification.

Scope. The multi-group coordinate-based meta-regression model from `imag.a.1057.pdf` has a different parameterization and is developed in the separate `imag_a.1057_appendix.pdf`. Its moment-matched aggregate and study-clustered NB likelihoods are not substituted for the independent-cell NB working likelihood here.

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1 Paper correspondence, model setup, and dimensions

1.1 A map to the supplied manuscript

Manuscript location	Material developed here
Section 2.1; (2.1)–(2.4)	Fixed spline or QMC spatial basis
Section 2.2; (2.5)–(2.7)	Mass-univariate comparator and spatial-effect design
(2.8)–(2.10); S1.2–S1.3	Poisson/NB working likelihoods, binary specialization, and distinction from a normalized binary model
(2.11); Algorithm 1, Stage 1	Identifiable separable Poisson initialization
(2.12)–(2.13); (S2.36)–(S2.37)	Family-specific scoring and exact versus approximate Kronecker preconditioning
(2.14); (S2.38)–(S2.42)	Subject-centered Taylor expansion, dimensions, and correct weighted basis product
(2.15)–(2.18)	Voxel contrasts, Fisher information, and a matching subject-level sandwich
S1.1; (S1.19)–(S1.23)	Logistic and Firth comparator derivatives
S2.2; (S2.43)–(S2.50)	Weighted-design QR and triangular sandwich solves
S2.3	Toy slope-map variance, coefficient permutation, and corrected subject/spatial moment factors

Equation numbers in the left column refer to the supplied manuscript; numbers elsewhere in this appendix refer to the present document. The manuscript sometimes labels a full-model likelihood by γ ; we use β for its full spatial coefficient vector and reserve γ for the separable initialization’s global coefficients, except in explicitly labelled local supplementary conventions.

1.2 Notation crosswalk and deliberate disambiguations

The main-text conventions are retained rather than importing the different conventions of `imag.a.1057.pdf`. In particular, here Z_i and B_j are *columns*; the rows of the two design matrices are $Z_i^\top$ and $B_j^\top$.

Manuscript symbol	Meaning and use in this appendix
M, N, R, P	Numbers of subjects, voxels, covariates, and spatial bases
Z, B, X	Designs of dimensions $M \times R$, $N \times P$, and $MN \times RP$, with $X = Z \otimes B$
Y, μ, η	Subject-major stacked count, mean, and log-mean vectors of length MN
$\tilde{Y}, \tilde{\beta}$	The paper’s $M \times N$ data array and $R \times P$ coefficient reshape from (2.14)
φ_j	Main-text (2.5)’s R -vector of voxelwise effects; in the spatial model $\varphi_j = (I_R \otimes B_j^\top)\beta$
ξ, γ	Separable initialization coefficients of dimensions P and R , not the full RP -vector β
μ^B, μ^Z, W_B, W_Z	Separable spatial/study factors and their diagonal Poisson weight matrices in (2.13)
$\tilde{Z}, \tilde{\eta}, \tilde{B}$	Centered covariates, subject-average log mean, and singly weighted basis from (2.14)
$\alpha, I(\beta)$	Global NB dispersion and regression Fisher information
C, S, W_j	Contrast $C \in \mathbb{R}^{S \times R}$, number of tested constraints, and the <i>signed</i> statistic in (2.16)
$\bar{\eta}_C, \bar{\mu}_C$	Contrast maps $(C \otimes B)\beta$ and their componentwise exponentials, not subject averages

New helpers are explicitly introduced: `vecrow` fixes the paper’s vectorization order; $\tilde{\eta}$ and $\tilde{\mu}$ are $M \times N$ arrays; $H = -\nabla^2 \ell$ is observed information; P_0 is a fixed full-model preconditioner; and Q_j names a squared Wald statistic, leaving W_j signed. We use the auxiliary NB shape $\nu = 1/\alpha$ so that r remains the covariate index.

Several manuscript symbols have different local meanings. Section 2.1 uses M for the number of Fourier features; we call that number P_f to distinguish it from subjects. Its QMC coefficients, also called γ there, are not the Stage 1 global coefficients. Supplement S1.1 calls its $M \times R$ logistic design X and its voxelwise coefficient β_j ; our comparator uses the main-text names Z and φ_j , with this mapping stated again in Section 13. Supplement S2.2 reuses B, M, R, S, Z for sandwich and QR quantities; we distinguish those from the spatial basis, dimensions, contrasts, and study design below. Supplement S2.3 reverses the Kronecker order and reuses α as an intercept; its local convention is translated explicitly in Section 12.

1.3 Data and stacking convention

There are M independent subjects and N voxels. Subject i supplies a binary image $Y_i = (Y_{i1}, \dots, Y_{iN})^\top$, $Y_{ij} \in \{0, 1\}$. Independence *within* an image is a working-likelihood assumption, not a consequence of smoothing its mean. Let $\tilde{Y} \in \mathbb{R}^{M \times N}$ store the images by row. Stack subjects slowest: $Y = (Y_1^\top, \dots, Y_M^\top)^\top = \text{vec}_{\text{row}}(\tilde{Y})$. For any matrix, vec_{row} stacks rows in order, equivalently applying ordinary column-vectorization to its transpose.

The fixed subject design is $Z \in \mathbb{R}^{M \times R}$ and the fixed spatial basis is $B \in \mathbb{R}^{N \times P}$. Rows are $Z_i^\top$ and $B_j^\top$; $Z_i \in \mathbb{R}^R$ and $B_j \in \mathbb{R}^P$ are columns. Stack the R spatial maps as

$$\beta = (\beta_1^\top, \dots, \beta_R^\top)^\top \in \mathbb{R}^{RP}, \quad \tilde{\beta} = \begin{pmatrix} \beta_1^\top \\ \vdots \\ \beta_R^\top \end{pmatrix}, \quad \beta = \text{vec}_{\text{row}}(\tilde{\beta}).$$

The model corresponding to manuscript (2.6)–(2.7) is

$$\eta_{ij} = \log \mu_{ij} = \sum_{r=1}^R Z_{ir} B_j^\top \beta_r = (Z_i^\top \otimes B_j^\top) \beta, \quad (1.1)$$

$$\tilde{\eta} = Z \tilde{\beta} B^\top \in \mathbb{R}^{M \times N}, \quad (1.2)$$

$$\eta = (Z \otimes B) \beta = X \beta, \quad X \in \mathbb{R}^{MN \times RP}. \quad (1.3)$$

Here $\tilde{\eta}$ is the *array of log means*, not a Hessian. The study design is $X_i = Z_i^\top \otimes B \in \mathbb{R}^{N \times RP}$. Let $x_{ij} = Z_i \otimes B_j$ be a cell design column. The number of regression coefficients is $d = RP$.

The coefficient of covariate Z_{ir} at voxel j is $\varphi_{jr} = B_j^\top \beta_r$, component r of the paper's φ_j . Thus $\partial \eta_{ij} / \partial Z_{ir} = \varphi_{jr}$. An intercept column of Z supplies a baseline map, not an additional constant global covariate effect. A common smooth basis reduces the number of coefficients but does not itself specify residual spatial covariance between Y_{ij} and Y_{ik} .

1.4 Derive the coefficient order rather than assuming a vec convention

The cell (i, j) has stacked row index $(i-1)N + j$, and the coefficient (r, p) has index $(r-1)P + p$. By the definition of a Kronecker product,

$$(Z \otimes B)_{(i-1)N+j, (r-1)P+p} = Z_{ir} B_{jp}.$$

Multiplication by $\beta = \text{vec}_{\text{row}}(\tilde{\beta})$ therefore gives $\sum_{r,p} Z_{ir} \tilde{\beta}_{rp} B_{jp}$, which is exactly $(Z \tilde{\beta} B^\top)_{ij}$. This proves both (2.7)'s subject-major stacking and the paper's $R \times P$ reshape convention.

More generally, if A is $R \times R$, K is $P \times P$, and D is $R \times P$, then

$$(ADK)_{rp} = \sum_{q,l} A_{rq} D_{ql} K_{lp}.$$

The coefficient multiplying D_{ql} is $(A \otimes K^\top)_{(r-1)P+p, (q-1)P+l}$, proving

$$\text{vec}_{\text{row}}(ADK) = (A \otimes K^\top) \text{vec}_{\text{row}}(D).$$

Ordinary column-vectorization of that same $R \times P$ array would interchange the coefficient order. It is valid only with the corresponding permutation of the design and covariance.

1.5 Spline and QMC bases are fixed design choices

For coordinates $s_j = (x_j, y_j, z_j)^\top$ a tensor spline column is indexed by (a, b, c) :

$$B_{j,(a,b,c)} = B_a^{(x)}(x_j)B_b^{(y)}(y_j)B_c^{(z)}(z_j).$$

Then $B_j^\top \beta_r$ is the triple sum of these products times the corresponding map coefficients, recovering (2.1)–(2.2). This uses the paper’s uppercase univariate basis functions. Its generic voxel index i in Section 2.1 is changed to j here, so that i consistently indexes subjects in the regression. Its generic spline coefficient ω_{abc} becomes the corresponding component of β_r for covariate map r . On a full rectangular grid the full basis can itself be built by Kronecker products. Masking voxels or dropping weakly supported columns does not preserve every such full-grid factorization. This spatial tensor construction is distinct from $X = Z \otimes B$.

For P_f fixed Fourier features, the manuscript’s alternative has

$$B_{jk} = \sqrt{2/P_f} \cos(\omega_k^\top s_j + b_k), \quad k = 1, \dots, P_f.$$

Thus $B_j = \varphi_{\text{QMC}}(s_j)$ before any additional intercept column is appended. The same feature vector in (2.3) is multiplied by a generic coefficient vector; the spatial GLM supplies one such vector β_r for each covariate. The frequencies and phases are generated before regression and held fixed; an added intercept, if independent, increases P by one. For clarity P_f counts features, avoiding the manuscript Section 2.1 reuse of M , which here always counts subjects. Conditioning on this basis, every regression derivative below is identical for splines and Fourier features. No differentiation through a chosen knot grid or QMC sequence is involved.

An intercept should not be appended blindly: a complete partition-of-unity spline basis already spans the constant function. If $Bc_B = \mathbf{1}_N$, adding another constant column makes the basis rank deficient unless a redundant direction is removed. For full-column-rank Z, B , $\text{rank}(Z \otimes B) = \text{rank}(Z) \text{rank}(B) = RP$. Having a constant direction in both factors does not itself duplicate the intercept in the *tensor-product* design. The additive initialization has a different intercept problem, derived in Section 6.

2 Binary observations: count working models versus normalized binary models

2.1 What conditioning a Poisson count on zero or one changes

For $Y \sim \text{Poisson}(\lambda)$, $P(Y = 0) = e^{-\lambda}$ and $P(Y = 1) = \lambda e^{-\lambda}$. After explicitly conditioning on $Y \in \{0, 1\}$,

$$P(Y = 1 \mid Y \leq 1) = \frac{\lambda}{1 + \lambda}, \quad P(Y = 0 \mid Y \leq 1) = \frac{1}{1 + \lambda}.$$

With $\lambda = e^\eta$, the conditional probability is logistic: $p = e^\eta / (1 + e^\eta)$. This is the exact conditioning calculation underlying manuscript S1.2. But observing only zeros and ones is not itself an instruction to condition the count likelihood. For an observed $y \in \{0, 1\}$ the two log-likelihoods are

$$\ell_{\text{count}}(\eta) = y\eta - e^\eta, \tag{2.1}$$

$$\ell_{\text{conditional}}(\eta) = \ell_{\text{count}}(\eta) - \log P_\eta(Y \leq 1) \tag{2.2}$$

$$= y\eta - e^\eta - \{-e^\eta + \log(1 + e^\eta)\} \tag{2.3}$$

$$= y\eta - \log(1 + e^\eta). \tag{2.4}$$

Their difference, $e^\eta - \log(1 + e^\eta)$, is parameter-dependent. Consequently their scores and Hessians differ. The manuscript's (2.8) uses the unconditioned count objective; it is not made identical to logistic regression by restricting the observed dataset to zero and one.

For small λ , $\lambda/(1 + \lambda) = \lambda - \lambda^2 + O(\lambda^3)$ and $\log(1 + \lambda) = \lambda - \lambda^2/2 + O(\lambda^3)$. This gives a low-rate approximation, not equality of the fitted objectives. Under the unconditioned count model, $\mu = e^\eta$ is an intensity and is not constrained below one. If it is interpreted as a Bernoulli mean, fitted values must satisfy that additional scientific requirement; clipping them after fitting changes neither the fitted likelihood nor its derivative correctly.

2.2 The same issue for NB evaluated on binary data

For an NB count with mean μ and dispersion $\alpha > 0$,

$$P(Y = 0) = (1 + \alpha\mu)^{-1/\alpha}, \quad \frac{P(Y = 1)}{P(Y = 0)} = \frac{\mu}{1 + \alpha\mu}.$$

Normalizing these two masses would give

$$p_{\text{conditional}} = \frac{\mu}{1 + (1 + \alpha)\mu}, \quad \text{logit}(p_{\text{conditional}}) = \log \mu - \log(1 + \alpha\mu). \quad (2.5)$$

That is a different binary model, with a nonlinear predictor in $\eta = \log \mu$. Its normalization must not be silently added to the NB working likelihood in manuscript (2.10). The following NB derivatives refer to that *unconditioned count* working likelihood evaluated on binary observations.

If the true conditional data are Bernoulli with mean p , $\text{Var}(Y | Z) = p(1 - p) < p$ for $0 < p < 1$. Thus $\mu + \alpha\mu^2$ with $\alpha > 0$ cannot be the literal conditional variance of a binary response with the same mean. Pooled heterogeneity or within-subject spatial dependence can still make a working independent model inadequate, but these are not positive Bernoulli conditional dispersion parameters. The working-model covariance and a subject-robust covariance therefore need separate interpretations.

3 A common chain rule for spatial coefficient maps

3.1 Differentiate the predictor before the likelihood

For coordinate $a = (r - 1)P + p$,

$$\frac{\partial \eta_{ij}}{\partial \beta_a} = Z_{ir} B_{jp} = x_{ij,a}, \quad \frac{\partial^2 \eta_{ij}}{\partial \beta_a \partial \beta_b} = 0.$$

Because $\mu_{ij} = e^{\eta_{ij}}$,

$$\partial_{\beta_a} \mu_{ij} = \mu_{ij} x_{ij,a}, \quad \partial_{\beta_a \beta_b}^2 \mu_{ij} = \mu_{ij} x_{ij,a} x_{ij,b}.$$

For a scalar likelihood $f(\mu)$, this means

$$f_\eta = \mu f_\mu, \quad f_{\eta\eta} = \mu^2 f_{\mu\mu} + \mu f_\mu.$$

The final term is essential: the log predictor is affine, but the mean itself is not.

3.2 Score, log-likelihood Hessian, and observed information

Write the working log-likelihood as $\ell(\beta) = \sum_{i,j} f_{ij}(\eta_{ij})$. Set $s_{ij} = f'_{ij}$ and $w_{ij} = -f''_{ij}$. Differentiating first and second gives

$$U_a = \partial_{\beta_a} \ell = \sum_{i,j} s_{ij} x_{ij,a}, \quad (3.1)$$

$$\partial_{\beta_b} U_a = \sum_{i,j} \{f''_{ij} x_{ij,b} x_{ij,a} + s_{ij} \partial_{\beta_b} x_{ij,a}\} \quad (3.2)$$

$$= - \sum_{i,j} w_{ij} x_{ij,a} x_{ij,b}. \quad (3.3)$$

Define $H = -\nabla_{\beta}^2 \ell$ and $I = E_{\beta} H$. Then

$$U = \sum_i X_i^{\top} s_i, \quad (3.4)$$

$$H = \sum_i X_i^{\top} \text{diag}(w_i) X_i, \quad (3.5)$$

$$H_{\beta_r \beta_{r'}} = \sum_i Z_{ir} Z_{ir'}^{\top} B^{\top} \text{diag}(w_i) B, \quad (3.6)$$

$$H_{(r,p),(r',p')} = \sum_i \sum_j Z_{ir} Z_{ir'}^{\top} B_{jp} B_{jp'}^{\top} w_{ij}. \quad (3.7)$$

The actual *log-likelihood Hessian* is the negative of every displayed information block. The scalar weights determine the observation family; the spatial projection is unchanged.

For implementation, compute $\tilde{B}_{jp} = w_{ij} B_{jp}$ and $B^{\top} \tilde{B}$ rather than constructing an $N \times N$ diagonal matrix. Assemble studies or batches without building X . A dense weighted basis product costs $O(NP^2)$ and uses an $N \times P$ workspace. Dense $RP \times RP$ information still costs $O((RP)^2)$ storage; avoiding the large design does not remove that cost.

3.3 An exact matrix-free Hessian–vector product

For a direction $v \in \mathbb{R}^{RP}$, let $\tilde{v} \in \mathbb{R}^{R \times P}$ satisfy $v = \text{vec}_{\text{row}}(\tilde{v})$. The directional change in the log mean has array

$$\tilde{q} = Z \tilde{v} B^{\top}, \quad \tilde{q}_{ij} = x_{ij}^{\top} v.$$

Multiply (3.7)’s full sum by v :

$$Hv = \sum_{i,j} w_{ij} x_{ij} (x_{ij}^{\top} v) \quad (3.8)$$

$$= \text{vec}_{\text{row}} \left[Z^{\top} \{ \tilde{w} \odot (Z \tilde{v} B^{\top}) \} B \right], \quad \tilde{w} = (w_{ij})_{i,j}. \quad (3.9)$$

The symbol $\odot$ is elementwise multiplication; $\tilde{w}$ is an array, whereas the paper’s W is the stacked diagonal matrix. This identity is exact for observed Poisson or NB regression curvature at fixed dispersion. Substituting Fisher weights gives Iv instead. Its quadratic form is $v^{\top} H v = \sum_{i,j} w_{ij} (x_{ij}^{\top} v)^2$, displaying symmetry and positive semidefiniteness without forming any dense Hessian.

Subject/voxel batches can accumulate the projected products without storing X , H , or all of $\tilde{q}$. With the indicated multiplication order, one dense product costs $O(MRP + MNP)$, aside from obtaining the current weights. For identified SPD information, an iterative linear solver with these products and the separable preconditioner can compute a direction or selected covariance column without dense information storage. This changes the numerical solve, not the likelihood. Residual convergence and information rank still matter; the preconditioner alone is not the final covariance.

4 Poisson working likelihood: scalar and spatial Hessian derivations

4.1 From the mass function to the second derivative

The mass function and log-likelihood are

$$P(Y = y) = \frac{e^{-\mu} \mu^y}{y!}, \quad f(\eta) = y\eta - e^\eta - \log \Gamma(y + 1).$$

When $y = 0$ or 1 , $\log \Gamma(y + 1) = 0$, giving (2.8). Differentiate at fixed data:

$$f_\eta = y - e^\eta = y - \mu, \quad (4.1)$$

$$f_{\eta\eta} = -e^\eta = -\mu, \quad (4.2)$$

$$U = \sum_{i,j} (Z_i \otimes B_j)(Y_{ij} - \mu_{ij}), \quad (4.3)$$

$$H = \sum_i (Z_i Z_i^\top) \otimes \{B^\top \text{diag}(\mu_i) B\}. \quad (4.4)$$

No count appears explicitly in H at a fixed parameter value. $H(\hat{\beta})$ still depends on the data through the fitted coefficient. The count-model expectation gives $I = H$.

Under the correctly specified independent Poisson model,

$$\text{Var}(U) = \sum_{i,j} x_{ij} x_{ij}^\top \text{Var}(Y_{ij}) = \sum_{i,j} \mu_{ij} x_{ij} x_{ij}^\top = I.$$

Under actual binary or spatially correlated data, the algebraic Hessian is unchanged but this information equality need not hold.

4.2 Exact score as a small projected matrix

Let $\tilde{\mu}_{ij} = \mu_{ij}$ and $R_Y = \tilde{Y} - \tilde{\mu}$. For the (r, p) coefficient the score is $\sum_i Z_{ir} \sum_j (R_Y)_{ij} B_{jp}$. This is entry (r, p) of $Z^\top R_Y B$, so

$$U = \text{vec}_{\text{row}}\{Z^\top (\tilde{Y} - \tilde{\mu}) B\}. \quad (4.5)$$

In particular $X^\top Y = \text{vec}_{\text{row}}(Z^\top \tilde{Y} B)$. This equality fixes the vectorization convention left implicit in some of the manuscript's formulas; using column-wise vec of the $R \times P$ matrix would order the coefficients differently. The count projection can be precomputed for a fixed design, while the mean projection must be updated at each exact score evaluation.

5 Independent NB working likelihood: binary simplification and derivatives

5.1 General NB2 likelihood and exact binary cancellation

Let $\nu = \alpha^{-1}$, with the manuscript's single dispersion $\alpha > 0$ shared over subjects and voxels. The NB2 mass is

$$P(Y = y) = \frac{\Gamma(y + \nu)}{\Gamma(\nu)\Gamma(y + 1)} \left(\frac{\nu}{\nu + \mu}\right)^\nu \left(\frac{\mu}{\nu + \mu}\right)^y.$$

Its mean and variance under that count model are μ and $\mu + \alpha\mu^2$. The scalar log-likelihood is

$$f = \log \Gamma(y + \nu) - \log \Gamma(\nu) - \log \Gamma(y + 1) + \nu \log \nu + y\eta - (y + \nu) \log(\nu + e^\eta). \quad (5.1)$$

For binary y , $\log \Gamma(y + \nu) - \log \Gamma(\nu) = y \log \nu$. Also $\log \Gamma(y + 1) = 0$ and $\log(\nu + \mu) = \log \nu + \log(1 + \alpha\mu)$. Substituting each term,

$$f = y \log \nu + \nu \log \nu + y\eta - (y + \nu)\{\log \nu + \log(1 + \alpha\mu)\} \quad (5.2)$$

$$= y\eta - (y + \alpha^{-1}) \log(1 + \alpha\mu) \quad (5.3)$$

$$= y \log \left(\frac{\mu}{1 + \alpha\mu} \right) - \alpha^{-1} \log(1 + \alpha\mu). \quad (5.4)$$

This derives manuscript (2.10) and (S1.34), including the cancellations. It is an exact simplification of the NB *count mass* at binary observations, not a change in the normalization of that mass.

5.2 First derivative at fixed original dispersion

Since $\partial_\eta \mu = \mu$,

$$f_\eta = y - (y + \alpha^{-1}) \frac{\alpha\mu}{1 + \alpha\mu} \quad (5.5)$$

$$= \frac{y(1 + \alpha\mu) - (1 + \alpha y)\mu}{1 + \alpha\mu} \quad (5.6)$$

$$= \frac{y - \mu}{1 + \alpha\mu}. \quad (5.7)$$

The denominator is present even for binary y and even though the link is logarithmic.

5.3 Quotient rule for observed curvature

Differentiate numerator and denominator separately:

$$f_{\eta\eta} = \frac{(-\mu)(1 + \alpha\mu) - (y - \mu)\alpha\mu}{(1 + \alpha\mu)^2} \quad (5.8)$$

$$= -\frac{\mu + \alpha\mu^2 + \alpha\mu y - \alpha\mu^2}{(1 + \alpha\mu)^2} \quad (5.9)$$

$$= -\frac{\mu(1 + \alpha y)}{(1 + \alpha\mu)^2}. \quad (5.10)$$

Thus

$$w_{ij}^{\text{obs}} = \frac{\mu_{ij}(1 + \alpha Y_{ij})}{(1 + \alpha\mu_{ij})^2}, \quad (5.11)$$

$$H_{\beta_r \beta_{r'}} = \sum_i Z_{ir} Z_{ir'} B^\top \text{diag}(w_i^{\text{obs}}) B, \quad (5.12)$$

$$U_{\beta_r} = \sum_i Z_{ir} B^\top \left\{ \frac{Y_i - \mu_i}{1 + \alpha\mu_i} \right\}. \quad (5.13)$$

Vector division is elementwise. These are independent-cell NB derivatives; there is no aggregation across subjects and no shared gamma multiplier inducing a rank-one voxel correction. For fixed positive α , $w^{\text{obs}} > 0$. Consequently the negative log-likelihood is convex in the affine regression coefficients and strictly convex when X has full column rank. Existence of a finite optimum and joint convexity with dispersion are separate questions.

5.4 Expected information and what it assumes

Taking an expectation under the NB working distribution gives

$$w^F = \frac{\mu(1 + \alpha \mathbb{E} Y)}{(1 + \alpha \mu)^2} = \frac{\mu}{1 + \alpha \mu}, \quad (5.14)$$

$$I = \sum_i X_i^\top \text{diag} \left(\frac{\mu_i}{1 + \alpha \mu_i} \right) X_i. \quad (5.15)$$

These are manuscript (2.17)'s weights. Independently,

$$\text{Var}_{\text{NB}}(f_\eta) = \frac{\mu + \alpha \mu^2}{(1 + \alpha \mu)^2} = \frac{\mu}{1 + \alpha \mu}.$$

The expectation of the observed weight uses only the mean, so it also gives the expected sensitivity of this working score under another distribution with the same mean. The variance identity does *not* then follow automatically. For a truly independent Bernoulli cell with mean $\mu < 1$,

$$\text{Var}_{\text{Bernoulli}}(f_\eta) = \frac{\mu(1 - \mu)}{(1 + \alpha \mu)^2},$$

and a whole subject may have additional covariance between its cells. This is precisely why sensitivity and score variability must be separated in a robust calculation.

5.5 Dispersion derivatives, transformations, and the boundary

Using the general NB likelihood, define $\psi(t) = d \log \Gamma(t)/dt$ and $\psi_1(t) = d\psi(t)/dt$. At fixed μ ,

$$f_\nu = \psi(y + \nu) - \psi(\nu) + \log \nu + 1 - \log(\nu + \mu) - \frac{y + \nu}{\nu + \mu}, \quad (5.16)$$

$$f_{\nu\nu} = \psi_1(y + \nu) - \psi_1(\nu) + \frac{1}{\nu} - \frac{1}{\nu + \mu} - \frac{\mu - y}{(\nu + \mu)^2}, \quad (5.17)$$

$$f_{\eta\nu} = \partial_\nu \left\{ \frac{\nu(y - \mu)}{\nu + \mu} \right\} = \frac{\mu(y - \mu)}{(\nu + \mu)^2}. \quad (5.18)$$

For the binary-specialized likelihood (5.4), one can also avoid the gamma functions entirely. Put $h = 1 + \alpha \mu$:

$$f_\alpha = \alpha^{-2} \log h - (y + \alpha^{-1})\mu/h, \quad (5.19)$$

$$f_{\alpha\alpha} = -2\alpha^{-3} \log h + 2\alpha^{-2}\mu/h + (y + \alpha^{-1})\mu^2/h^2, \quad (5.20)$$

$$f_{\eta\alpha} = -\mu(y - \mu)/h^2. \quad (5.21)$$

For example the two $\alpha^{-2}\mu/h$ terms in the second derivative come from differentiating $\log h$ in the first term and α^{-1} in the second. These formulas agree with $\nu = 1/\alpha$ and the full chain rule. Assemble $H_{\beta\alpha} = -\sum_{i,j} x_{ij} f_{\eta\alpha}$ and $H_{\alpha\alpha} = -\sum_{i,j} f_{\alpha\alpha}$.

For $\delta = \log \alpha$, differentiation gives

$$f_\delta = \alpha f_\alpha, \quad f_{\delta\delta} = \alpha^2 f_{\alpha\alpha} + \alpha f_\alpha, \quad H_{\delta\delta} = \alpha^2 H_{\alpha\alpha} - \alpha U_\alpha.$$

The extra score term is not zero away from an interior stationary point. Holding $\hat{\alpha}$ fixed while calculating (5.13) does not propagate dispersion uncertainty. The Poisson limit $\alpha \downarrow 0$ recovers $s = y - \mu$ and $w^{\text{obs}} = w^F = \mu$, but $\alpha = 0$ is a boundary, not an interior value of δ . Joint Wald inference at that boundary is not justified by the preceding interior derivatives.

6 The manuscript's separable initialization, not its final model

6.1 Additive predictor and identification

Stage 1 of manuscript Algorithm 1 uses a Poisson fit with

$$\eta_{ij}^{(0)} = B_j^\top \xi + Z_i^\top \gamma, \quad \mu_j^B = e^{B_j^\top \xi}, \quad \mu_i^Z = e^{Z_i^\top \gamma}, \quad \mu_{ij}^{(0)} = \mu_i^Z \mu_j^B.$$

These are the manuscript's μ^B and μ^Z , not the μ^X factors used in the other paper. The dimensionally explicit version of (2.11) is

$$\eta^{(0)} = (\mathbf{1}_M \otimes B)\xi + (Z \otimes \mathbf{1}_N)\gamma.$$

The unstacked components $\eta_B = B\xi$ and $\eta_Z = Z\gamma$ have lengths N and M and must be broadcast this way before they can be added. This has $P + R$ nominal coefficients, not PR . Here γ is globally constant across voxels and is only the initialization parameterization; the final β_r are spatial maps.

If $Bc_B = \mathbf{1}_N$ and $Zc_Z = \mathbf{1}_M$, the transformation $\xi \mapsto \xi + tc_B$, $\gamma \mapsto \gamma - tc_Z$ leaves every $\eta_{ij}^{(0)}$ unchanged. Therefore an unconstrained additive fit with both intercepts is unidentified. Remove the global intercept for Stage 1 or impose an explicit equivalent constraint. Centering the nonconstant covariates does not remove a separately included global intercept.

6.2 Exact compression into two count marginals

Introduce auxiliary count margins $Y_B = \tilde{Y}^\top \mathbf{1}_M$ and $Y_Z = \tilde{Y} \mathbf{1}_N$, of lengths N and M . Thus $(Y_B)_j = \sum_i Y_{ij}$ and $(Y_Z)_i = \sum_j Y_{ij}$. Define $h_B = \sum_j \mu_j^B$ and $h_Z = \sum_i \mu_i^Z$. Substituting the additive predictor into the full Poisson objective,

$$\ell_0 = \sum_{i,j} Y_{ij} (B_j^\top \xi + Z_i^\top \gamma) - \sum_{i,j} \mu_i^Z \mu_j^B + c(Y) \quad (6.1)$$

$$= Y_B^\top B\xi + Y_Z^\top Z\gamma - h_B h_Z + c(Y). \quad (6.2)$$

The two count marginals suffice for evaluating this likelihood. Fitting only $(Y_B)_j \sim \text{Poisson}(h_Z \mu_j^B)$ would drop $\sum_i (Y_Z)_i \log(\mu_i^Z / h_Z)$, which generally depends on γ . Exact compression is not the same as discarding the allocation of counts among subjects.

6.3 Scores and all initialization Hessian blocks

Set

$$\begin{aligned} W_B &= \text{diag}(\mu^B), & W_Z &= \text{diag}(\mu^Z), \\ a &= B^\top \mu^B, & t &= Z^\top \mu^Z, \\ K_B &= B^\top W_B B, & K_Z &= Z^\top W_Z Z. \end{aligned}$$

Exponential differentiation gives $\nabla_\xi h_B = a$, $\nabla_\xi^2 h_B = K_B$, $\nabla_\gamma h_Z = t$, and $\nabla_\gamma^2 h_Z = K_Z$. Differentiate (6.2) once:

$$U_\xi = B^\top Y_B - h_Z a, \quad U_\gamma = Z^\top Y_Z - h_B t.$$

Then differentiate the two scores:

$$-\partial_\xi U_\xi = h_Z K_B, \quad -\partial_\gamma U_\xi = at^\top, \quad (6.3)$$

$$-\partial_\gamma U_\gamma = h_B K_Z, \quad (6.4)$$

$$H_0 = \begin{pmatrix} h_Z K_B & at^\top \\ ta^\top & h_B K_Z \end{pmatrix}. \quad (6.5)$$

These formulas are written in nominal (ξ, γ) coordinates. If the global intercept is redundant, impose the chosen constraint before solving: for $(\xi^\top, \gamma^\top)^\top = J_0 \vartheta$ with fixed full-column-rank constraint map J_0 , use score $J_0^\top (U_\xi^\top, U_\gamma^\top)^\top$ and information $J_0^\top H_0 J_0$. Deleting a redundant coordinate is one such map. The constrained information is not the $PR \times PR$ information of the full spatially varying model at the same means.

6.4 Embedding a separable fit in the full coefficient maps

Suppose $Ze_0 = \mathbf{1}_M$ selects the full model's intercept column, and $Bc_B = \mathbf{1}_N$. Pad the identifiable Stage 1 global vector with zero in its removed intercept coordinate and call it $\hat{\gamma}$. Define

$$\tilde{\beta}^{(0)} = e_0 \hat{\xi}^\top + \hat{\gamma} c_B^\top, \quad \beta^{(0)} = \text{vec}_{\text{row}}(\tilde{\beta}^{(0)}). \quad (6.6)$$

Multiplication checks the initialization:

$$Z \tilde{\beta}^{(0)} B^\top = \mathbf{1}_M (B \hat{\xi})^\top + (Z \hat{\gamma}) \mathbf{1}_N^\top.$$

Thus it exactly reproduces the separable log means. The shorthand $\beta^{(0)} \leftarrow \hat{\xi}$ in Algorithm 1 cannot literally assign a P -vector to a PR -vector. Equation (6.6) supplies a dimensionally complete embedding that also retains the fitted global effects. If the required intercept directions are not spanned, an explicit projection is needed and the embedding is no longer exact. Stage 1 is Poisson, so it does not itself estimate an NB dispersion; Stage 2 must specify a positive starting α or a separate dispersion fit rather than inherit a nonexistent Poisson estimate.

7 Exact scoring and the Kronecker preconditioner

7.1 Newton and Fisher directions with the correct family score

A Taylor expansion of the score equation gives $0 \approx U(\beta) - H(\beta)\Delta$, hence the Newton direction $\Delta = H^{-1}U$. Fisher scoring uses $\Delta = I^{-1}U$ instead. The two family scores are

$$U_P = X^\top (Y - \mu), \quad U_{NB} = X^\top \text{diag}\{(1 + \alpha\mu)^{-1}\} (Y - \mu).$$

For either family define the IRLS working response $z^* = \eta + (Y - \mu)/\mu$. For Poisson $W = \text{diag}(\mu)$ and for NB $W = \text{diag}\{\mu/(1 + \alpha\mu)\}$. Then

$$X^\top W(z^* - \eta) = \begin{cases} X^\top (Y - \mu), & \text{Poisson,} \\ X^\top \{(Y - \mu)/(1 + \alpha\mu)\}, & \text{NB.} \end{cases}$$

Solving $X^\top W X \Delta = X^\top W(z^* - \eta)$ therefore reproduces the family-specific Fisher direction. The numerator in manuscript (2.12) and Algorithm 1 is correct for Poisson, but must carry the NB residual denominator if it is intended as NB likelihood Fisher scoring.

7.2 Deriving the exact factorization at separable Poisson means

At $\mu_{ij}^{(0)} = \mu_i^Z \mu_j^B$, the stacked diagonal weights satisfy $W_0 = W_Z \otimes W_B = \text{diag}(\mu^Z) \otimes \text{diag}(\mu^B)$. Using $(A \otimes B)(C \otimes D) = AC \otimes BD$ twice,

$$P_0 = X^\top W_0 X = (Z^\top \otimes B^\top) \{W_Z \otimes W_B\} (Z \otimes B) \quad (7.1)$$

$$= \{Z^\top W_Z Z\} \otimes \{B^\top W_B B\} \quad (7.2)$$

$$= K_Z \otimes K_B. \quad (7.3)$$

If both factors are nonsingular, multiplication proves $P_0^{-1} = K_Z^{-1} \otimes K_B^{-1}$. This is the PR -dimensional full-model Poisson information *evaluated at* the separable means, not H_0 above.

When later means cease to be separable, it is a fixed preconditioner, not the current exact information.

One can solve $P_0 d = g$ without forming the Kronecker matrix. Reshape $g = \text{vec}(G)$ and $d = \text{vec}(D)$ with $G, D \in \mathbb{R}^{P \times R}$, so their column stacking matches the map-major β order. Then

$$(K_Z \otimes K_B) \text{vec}(D) = \text{vec}(K_B D K_Z^\top) = \text{vec}(G).$$

Solve $K_B W = G$, followed by $K_Z D^\top = W^\top$. Factorization costs $O(P^3 + R^3)$, with $O(P^2 + R^2)$ factor storage, rather than a dense $O((PR)^3)$ factorization.

7.3 Why separable NB means do not give these exact weights

NB Fisher weights at the same means are $w_{ij} = \mu_i^Z \mu_j^B / (1 + \alpha \mu_i^Z \mu_j^B)$. For study factors a, b and voxel factors c, d , their determinant is

$$\frac{ac}{1 + \alpha ac} \frac{bd}{1 + \alpha bd} - \frac{ad}{1 + \alpha ad} \frac{bc}{1 + \alpha bc} \quad (7.4)$$

$$= \frac{\alpha abcd(a - b)(d - c)}{(1 + \alpha ac)(1 + \alpha ad)(1 + \alpha bc)(1 + \alpha bd)}. \quad (7.5)$$

It is generally nonzero, whereas a separable two-by-two weight grid has determinant zero. Thus manuscript (2.13)'s Poisson products are not an exact NB weight factorization. One may deliberately use P_0 as an SPD approximation for NB, but the NB score and final information must retain their own formulas.

7.4 What a fixed preconditioner can and cannot guarantee

For a maximized objective ℓ and $P_0 \succ 0$, the direction $d = P_0^{-1} U$ satisfies $U^\top d = U^\top P_0^{-1} U > 0$ unless $U = 0$. This is a local ascent direction. A full step of length one need not increase ℓ . For example, maximize $\ell(b) = -hb^2/2$ using scalar $P_0 = p > 0$:

$$b_{\text{new}} = b + p^{-1}(-hb) = (1 - h/p)b.$$

Convergence requires $|1 - h/p| < 1$, equivalently $0 < h/p < 2$. Positive p alone does not imply that inequality. Line search, damping, or other suitable optimization conditions are needed for a convergence guarantee. The same caveat applies before introducing a Taylor approximation to the score.

8 Subject-centered Taylor approximation: complete derivation

8.1 Average log mean and centered predictor

Define

$$\bar{z} = Z^\top \mathbf{1}_M / M, \quad Q_M = I_M - \mathbf{1}_M \mathbf{1}_M^\top / M, \quad \tilde{Z} = Q_M Z, \quad \bar{\eta} = B \tilde{\beta}^\top \bar{z}.$$

Here $\tilde{Z}$ has the paper's intended meaning: the centered $M \times R$ covariate matrix. The centering matrix Q_M alone is $M \times M$ and is not $\tilde{Z}$; the right multiplication by Z is needed in the definition printed after (2.14). Then

$$Z = \mathbf{1}_M \bar{z}^\top + \tilde{Z}, \quad \tilde{\eta} = \mathbf{1}_M \bar{\eta}^\top + \Delta, \quad \Delta = \tilde{Z} \tilde{\beta} B^\top.$$

For each voxel, $\sum_i \Delta_{ij} = 0$. Let $a_j = e^{\bar{\eta}_j}$ and $a = \exp(\bar{\eta})$. Importantly, $a_j = \exp(M^{-1} \sum_i \eta_{ij})$ is not generally the subject-average mean $M^{-1} \sum_i e^{\eta_{ij}}$. Jensen's inequality gives $a_j \leq M^{-1} \sum_i \mu_{ij}$.

8.2 Scalar Taylor expansion with a remainder

Because $\mu_{ij} = a_j e^{\Delta_{ij}}$, expand the scalar exponential:

$$\mu_{ij} = a_j(1 + \Delta_{ij}) + \mathcal{R}_{ij}, \quad (8.1)$$

$$\mathcal{R}_{ij} = \frac{1}{2} a_j e^{t_{ij}} \Delta_{ij}^2, \quad t_{ij} \text{ lies between } 0 \text{ and } \Delta_{ij}, \quad (8.2)$$

$$0 \leq \mathcal{R}_{ij} \leq \frac{1}{2} a_j e^{|\Delta_{ij}|} \Delta_{ij}^2. \quad (8.3)$$

In matrix form, with $D_a = \text{diag}(a)$,

$$\tilde{\mu}_{\text{lin}} = \mathbf{1}_M a^\top + \Delta D_a = \mathbf{1}_M a^\top + \tilde{Z} \tilde{\beta} B^\top D_a. \quad (8.4)$$

The multiplication by D_a weights columns (voxels), not subject rows. This is exactly the first-order expansion in manuscript S2.1 before the final matrix simplification. It can yield negative approximate means if $\Delta_{ij} < -1$. It should then not be treated as a valid new Poisson or NB mean inside a log-likelihood.

8.3 Projecting the approximate Poisson score

Start with (4.5) and substitute (8.4):

$$Z^\top (\tilde{Y} - \tilde{\mu}_{\text{lin}}) B = Z^\top \tilde{Y} B - (Z^\top \mathbf{1}_M)(a^\top B) \quad (8.5)$$

$$- (Z^\top \tilde{Z}) \tilde{\beta} (B^\top D_a B). \quad (8.6)$$

Every term has dimension $R \times P$. For $A \in \mathbb{R}^{R \times R}$ and $K \in \mathbb{R}^{P \times P}$,

$$\text{vec}_{\text{row}}(A \tilde{\beta} K) = (A \otimes K^\top) \text{vec}_{\text{row}}(\tilde{\beta}).$$

Here $K_a = B^\top D_a B$ is symmetric, so row vectorization gives

$$U_{\text{lin}} = \text{vec}_{\text{row}}(Z^\top \tilde{Y} B) - (Z^\top \mathbf{1}_M) \otimes (B^\top a) - \{(Z^\top \tilde{Z}) \otimes K_a\} \beta. \quad (8.7)$$

Also $Z^\top \tilde{Z} = \tilde{Z}^\top \tilde{Z}$ because $Q_M^\top Q_M = Q_M$. The dimensions are $RP \times 1$ for every term. The current $\bar{\eta}, a, K_a$ depend on the current β ; this is a residual approximation at an iterate, not an assertion that its derivative is just the final Kronecker factor.

8.4 Differentiate the moving expansion point as well

To derive that derivative, put $\bar{x}_j = \bar{z} \otimes B_j$. Then $\bar{\eta}_j = \bar{x}_j^\top \beta$ and $\Delta_{ij} = (x_{ij} - \bar{x}_j)^\top \beta$. Since $a_j = e^{\bar{\eta}_j}$, the product rule gives

$$\nabla_\beta \{a_j(1 + \Delta_{ij})\} = a_j(1 + \Delta_{ij}) \bar{x}_j + a_j(x_{ij} - \bar{x}_j) \quad (8.8)$$

$$= a_j(x_{ij} + \Delta_{ij} \bar{x}_j). \quad (8.9)$$

Therefore the negative Jacobian of the approximate score is

$$A_{\text{lin}} = -\frac{\partial U_{\text{lin}}}{\partial \beta^\top} = \sum_{i,j} a_j x_{ij} (x_{ij} + \Delta_{ij} \bar{x}_j)^\top. \quad (8.10)$$

The second term need not be symmetric, so U_{lin} is not in general the gradient of any scalar twice-differentiable objective. It should not be assigned a ‘‘Hessian’’ by retaining only $(Z^\top \tilde{Z}) \otimes K_a$.

For a direct check, take one voxel, two subject rows (1, −1) and (1, 1), and parameters (η_0, b) . With $a = e^{\eta_0}$ the approximate score is

$$U_{\text{lin}} = \begin{pmatrix} Y_1 + Y_2 - 2a \\ -Y_1 + Y_2 - 2ab \end{pmatrix}, \quad A_{\text{lin}} = \begin{pmatrix} 2a & 0 \\ 2ab & 2a \end{pmatrix}.$$

For $b \neq 0$ the mixed derivatives differ. If this approximate score defines the final estimator, its estimating-equation sensitivity is (8.10), not automatically the exact likelihood's information. A likelihood-based final covariance requires an exact-score refinement or justification that the approximation error is negligible on the relevant inferential scale. As with any working estimating equation, a sandwich still requires a well-defined target and appropriate centered score moments.

8.5 The weighted basis must appear once, not squared

Define $\tilde{B} = D_a B$, which has dimension $N \times P$. Then the last factor in (8.7) is

$$K_a = B^\top \tilde{B} = B^\top D_a B.$$

In contrast, $\tilde{B}^\top \tilde{B} = B^\top D_a^2 B$ would weight by a_j^2 , not a_j . The product $B^\top \tilde{B}$ printed in (2.14), Algorithm 1, and (S2.42) already has the correct single weight under $\tilde{B} = D_a B$. It should not be reinterpreted as $\tilde{B}^\top \tilde{B}$. An alternative square-root-weighted definition $\hat{B} = D_a^{1/2} B$ would permit $\hat{B}^\top \hat{B} = B^\top D_a B$. Neither weighted basis has dimension $N \times N$ unless $P = N$.

The fixed data projection remains $\text{vec}_{\text{row}}(Z^\top \tilde{Y} B)$, using the *unweighted* B ; weighting that data basis would change the score. Also the final $RP \times RP$ Kronecker factor multiplies the vector β , not its $R \times P$ reshape $\tilde{\beta}$. The latter belongs inside the matrix product in (8.6), followed by row-vectorization. These distinctions make (2.14)'s data projection and reshaped coefficient notation consistent with Algorithm 1 and (S2.42); they do not call for replacing the already correct $B^\top \tilde{B}$ factor.

8.6 Error in the score and a checkable small example

The exact and approximate Poisson scores differ by

$$U - U_{\text{lin}} = - \sum_{i,j} x_{ij} \mathcal{R}_{ij}.$$

For any compatible vector norm, triangle inequality and (8.3) yield

$$\|U - U_{\text{lin}}\| \leq \frac{1}{2} \sum_{i,j} \|x_{ij}\| a_j e^{|\Delta_{ij}|} \Delta_{ij}^2.$$

Large centered log effects invalidate the assumption that the first-order residual is close to the exact one, even if the mean lesion incidence itself is small.

For a single voxel with intercept log mean η_0 , two centered covariate values $(-1, 1)$, and coefficient b , let $a = e^{\eta_0}$. The exact means are (ae^{-b}, ae^b) ; the approximation gives $(a(1 - b), a(1 + b))$. The intercept mean projection is $2a \cosh b$ versus $2a$. The covariate mean projection is $2a \sinh b$ versus $2ab$. Thus the intercept error begins at ab^2 and the covariate error at $ab^3/3$. In the scalar basis case $B = 1$, the correct $K_a = a$, whereas squaring the weighted basis incorrectly gives a^2 . This example distinguishes both the intended approximation and the unintended weighting change.

8.7 An NB score approximation must also expand its denominator

The manuscript's Taylor residual is a Poisson calculation. For NB define the scalar score $s(\eta) = (y - e^\eta)/(1 + \alpha e^\eta)$. At $\eta = \bar{\eta}_j$,

$$s(\bar{\eta}_j) = \frac{y - a_j}{1 + \alpha a_j}, \quad s'(\bar{\eta}_j) = -\frac{a_j(1 + \alpha y)}{(1 + \alpha a_j)^2}.$$

A first-order approximation to the actual NB score is therefore

$$s_{ij} \approx \frac{Y_{ij} - a_j}{1 + \alpha a_j} - \frac{a_j(1 + \alpha Y_{ij})}{(1 + \alpha a_j)^2} \Delta_{ij}. \quad (8.11)$$

For a projected matrix expression set $k_j = (1 + \alpha a_j)^{-1}$ and $c_j = a_j/(1 + \alpha a_j)^2$. Then

$$S_{\text{lin}}^{\text{NB}} = (\tilde{Y} - \mathbf{1}_M a^\top) \text{diag}(k) - \Delta \text{diag}(c) - \alpha(\tilde{Y} \odot \Delta) \text{diag}(c), \quad (8.12)$$

$$U_{\text{lin}}^{\text{NB}} = \text{vec}_{\text{row}}(Z^\top S_{\text{lin}}^{\text{NB}} B). \quad (8.13)$$

The final data-dependent term is absent for Poisson. There is no automatic claim that it reduces to the same cached products or computational cost as (8.7). Its validity is a local Taylor approximation with fixed α , not the exact NB update in (2.12). As $\alpha \downarrow 0$ it reduces to the Poisson expansion.

9 Covariance for the spatially varying model

9.1 From a likelihood Taylor expansion to an inverse information

For an unpenalized fit $\hat{\beta}$, expand the score near its population target β_0 :

$$0 = U(\hat{\beta}) \approx U(\beta_0) - H(\beta_0)(\hat{\beta} - \beta_0).$$

Thus $\hat{\beta} - \beta_0 \approx H^{-1}U(\beta_0)$. Under a correctly specified regular likelihood, $\text{Var}(U) = I$ and the large-sample sensitivity is I , giving $\text{Cov}(\hat{\beta}) \approx I^{-1}$. An observed-information alternative uses $H(\hat{\beta})^{-1}$. These coincide algebraically for Poisson but not generally for NB. Neither is the exact finite-sample covariance of a nonlinear estimator.

A fixed initialization preconditioner is not substituted for $I(\hat{\beta})$ merely because it was used to optimize the fit. Inference must use the curvature or sensitivity of the final estimating equation, or explicitly quantify an approximation to it.

9.2 Selected covariance by a solve

For $C_\beta \in \mathbb{R}^{k \times RP}$, solve

$$IV = C_\beta^\top.$$

Then $V = I^{-1}C_\beta^\top$ and $C_\beta V = C_\beta I^{-1}C_\beta^\top$ is the desired $k \times k$ covariance. This needs k right-hand sides rather than all RP columns of an inverse. A coordinate variance uses a coordinate row; a voxelwise contrast uses the row derived in Section 11. Batch voxel contrasts to avoid storing a full $N \times N$ spatial covariance when only its diagonal is required.

9.3 Design-supported block structure

By (3.7), a spatial cross block is zero if $Z_{ir}Z_{ir'} = 0$ for every subject. Connect columns of Z when they co-occur in a row. Different connected components give independent information blocks after a coefficient permutation: $H = \text{blkdiag}(H_1, \dots, H_K)$. Direct multiplication proves $H^{-1} = \text{blkdiag}(H_1^{-1}, \dots, H_K^{-1})$ when each block is invertible.

An intercept column and continuously varying covariates usually connect the SV maps, so group-style independent spatial blocks are not generally available here. No direct edge is weaker than no connecting path: for $\begin{pmatrix} a & b & 0 \\ b & c & e \\ 0 & e & f \end{pmatrix}$ the $(1,3)$ inverse entry is $be/\det H$, generally nonzero. Accidental zero entries at one fit should not define reusable structural blocks.

9.4 Joint dispersion adjustment

If α is jointly estimated, write its observed information as

$$\mathcal{I}_{\text{obs}} = \begin{pmatrix} H_{\beta\beta} & h \\ h^\top & a_\alpha \end{pmatrix}, \quad h = H_{\beta\alpha}, \quad a_\alpha = H_{\alpha\alpha}.$$

For a system with right-hand side $(b, 0)$, the nuisance equation gives $x_\alpha = -a_\alpha^{-1}h^\top x_\beta$. Substitution into the regression equation gives

$$(H_{\beta\beta} - ha_\alpha^{-1}h^\top)x_\beta = b.$$

Hence the regression block of the joint inverse is

$$(\mathcal{I}_{\text{obs}}^{-1})_{\beta\beta} = (H_{\beta\beta} - ha_\alpha^{-1}h^\top)^{-1}. \quad (9.1)$$

This requires the stated inverses; positive-definite joint curvature gives a PSD information reduction and a nonnegative covariance adjustment relative to treating dispersion as known.

For the correctly specified independent NB count likelihood, $E f_{\eta\alpha} = 0$ because (5.21) is proportional to $Y - \mu$. Regression and dispersion are therefore orthogonal in its expected information, but the observed cross block need not vanish. Alternating regression and dispersion updates does not itself imply orthogonality or supply the joint adjustment. For robust inference, stack both score equations and form their joint sandwich; adjusting only the regression bread does not account for nuisance-score variability.

10 Subject-clustered sandwich: matching score, bread, and normalization

10.1 Derivation with summed rather than averaged quantities

Let U_i be the chosen family's subject score and $U = \sum_i U_i$. Define

$$A = -\sum_i \frac{\partial U_i}{\partial \beta^\top}, \quad J = \sum_i U_i U_i^\top.$$

At a suitable target, the score linearization gives $\hat{\beta} - \beta_0 \approx A^{-1} \sum_i U_i$. For independent subjects and valid moment assumptions, the empirical covariance is

$$V_{\text{robust}} = A^{-1} J A^{-\top}. \quad (10.1)$$

No further $1/M$ multiplies this formula: A and J are already sums. If $\bar{A} = A/M$ and $\bar{J} = J/M$, $(1/M)\bar{A}^{-1}\bar{J}\bar{A}^{-\top} = A^{-1}JA^{-\top}$. This checks the normalization explicitly.

For the likelihood score, $A = H$. The NB expected-sensitivity weights in (5.15) are an asymptotically justified alternative under the correct conditional mean and appropriate regularity. They are not a license to change the score to the Poisson residual.

10.2 The family-specific subject score

The correct scores are

$$U_i^{\text{P}} = X_i^\top (Y_i - \mu_i), \quad U_i^{\text{NB}} = X_i^\top \text{diag}\{(1 + \alpha\mu_i)^{-1}\}(Y_i - \mu_i).$$

The manuscript's (2.18) uses the Poisson form for both families. To interpret it as an NB likelihood sandwich, replace the score by the NB expression above and use its matching sensitivity. A common log link does not make the two likelihood scores identical.

Write $U_i = \sum_j u_{ij}$ and expand its outer product:

$$U_i U_i^\top = \sum_j u_{ij} u_{ij}^\top + \sum_{j \neq k} u_{ij} u_{ik}^\top.$$

Subject clustering retains the cross-voxel terms. A cellwise sum of outer products drops them and is not robust to arbitrary residual dependence within a lesion image. If subjects themselves are dependent, the independent unit must be enlarged appropriately; more voxels do not supply more independent subjects.

Consistency also needs the relevant mean and law-of-large-numbers conditions. Under a misspecified fixed design, uncentered score outer products with nonzero individual score means cannot automatically be equated with sampling variances. A sandwich estimates variability around the estimator's target; it does not remove mean misspecification or approximation bias.

10.3 Contrast sandwich as a Gram matrix of projected scores

Stack subject scores as columns, $\mathcal{U} = [U_1, \dots, U_M] \in \mathbb{R}^{RP \times M}$, so $J = \mathcal{U} \mathcal{U}^\top$. This is the column orientation of the matrix called U in manuscript (S2.48); the calligraphic symbol distinguishes it from the total score $U = \sum_i U_i$. For symmetric bread A , solve $AV = C_\beta^\top$ and let $G = \mathcal{U}^\top V$. Then

$$G^\top G = V^\top J V = C_\beta A^{-1} J A^{-1} C_\beta^\top.$$

This obtains the contrast covariance without constructing a dense meat or full inverse. Equivalently accumulate $g_i g_i^\top$ with $g_i = C_\beta A^{-1} U_i$. For a nonsymmetric estimating-equation bread solve $A^\top V = C_\beta^\top$ instead.

The Gram representation is PSD and has rank at most $\min(M, k, d)$. At an exactly solved unpenalized score equation, $\sum_i U_i = 0$, giving the stronger bound $\min(M - 1, k, d)$. A scalar degrees-of-freedom correction cannot increase that rank or compensate for an inadequate number of independent subjects.

10.4 Expand the manuscript's QR sandwich calculation

For Poisson, define the paper's weighted subject design $X_{w,i} = W_i^{1/2} X_i$, with $W_i = \text{diag}(\mu_i)$. The stacked weighted design X_w satisfies

$$X_w^\top X_w = \sum_i X_{w,i}^\top X_{w,i} = \sum_i X_i^\top W_i X_i = A.$$

This is (S2.44)–(S2.47). The supplement calls the bread B and the meat M ; we retain A and J to avoid confusing them with the spatial basis and number of subjects.

For full-column-rank X_w , take a thin QR factorization

$$X_w = \mathcal{Q} \mathcal{R}, \quad \mathcal{Q}^\top \mathcal{Q} = I_{RP}, \quad A = \mathcal{R}^\top \mathcal{R}.$$

Here $\mathcal{R}$ is *upper* triangular in the standard QR convention, not lower triangular as described after (S2.48). To compute $\mathcal{S}_U = A^{-1} \mathcal{U}$, solve

$$\mathcal{R}^\top T_U = \mathcal{U}, \quad \mathcal{R} \mathcal{S}_U = T_U.$$

Substitution gives $\mathcal{R}^\top \mathcal{R} \mathcal{S}_U = \mathcal{U}$, which verifies the solve order. The supplement names these intermediates Z and S ; $T_U, \mathcal{S}_U$ keep them separate from the study design and contrast count here. Now multiply:

$$A^{-1} J A^{-1} = A^{-1} \mathcal{U} \mathcal{U}^\top A^{-1} \tag{10.2}$$

$$= (A^{-1} \mathcal{U})(A^{-1} \mathcal{U})^\top = \mathcal{S}_U \mathcal{S}_U^\top. \tag{10.3}$$

This derives (S2.49)–(S2.50) without constructing a dense meat. For only k contrasts, the smaller solves in the previous subsection avoid storing all $RP \times M$ solved score columns.

For NB expected-sensitivity bread the same algebra uses $W_i = \text{diag}\{\mu_i/(1 + \alpha\mu_i)\}$ and the NB subject scores. For observed NB bread, use the observed weights instead. Changing the bread's weights does not eliminate the NB residual denominator in $\mathcal{U}$.

The full $MN \times RP$ stack need not be stored. If $\mathcal{R}_{\text{old}}$ represents already processed weighted rows, QR-factor the stack of $\mathcal{R}_{\text{old}}$ and the next block $X_{w,\text{next}}$. The resulting factor obeys

$$\mathcal{R}_{\text{new}}^\top \mathcal{R}_{\text{new}} = \mathcal{R}_{\text{old}}^\top \mathcal{R}_{\text{old}} + X_{w,\text{next}}^\top X_{w,\text{next}}.$$

This proves the blockwise accumulation of the same bread factor. With a thin SVD $X_w = Q_w D_w V_w^\top$, the equivalent solve is $A^{-1}\mathcal{U} = V_w D_w^{-2} V_w^\top \mathcal{U}$ when every singular value is positive. Since $\kappa_2(A) = \kappa_2(X_w)^2$, weighted-design QR/SVD can avoid the extra rounding error of explicitly forming a Gram matrix, but cannot create information in a singular design.

11 Voxelwise inference under the manuscript's spatial-effect setup

11.1 Linear map contrasts and dimension checks

Let $C \in \mathbb{R}^{S \times R}$ specify the tested spatial covariate effects. The map vector at voxel j is the paper's $\varphi_j = (I_R \otimes B_j^\top)\beta$. Across voxels, retain its contrast-map notation $\bar{\eta}_C = (C \otimes B)\beta$ and $\bar{\mu}_C = \exp(\bar{\eta}_C)$. The bar with subscript C labels these contrasts; it does not mean the subject averaging used to define $\bar{\eta}$ in the Taylor expansion. Multiplying by C gives

$$T_j = C \otimes B_j^\top \in \mathbb{R}^{S \times RP}, \quad \hat{\delta}_j = T_j \hat{\beta} - \delta_{0j}, \quad V_j = T_j V_\beta T_j^\top.$$

This supplies the dimensions for manuscript (2.15)–(2.16). For S independent estimable constraints and nonsingular consistent V_j , $Q_j = \hat{\delta}_j^\top V_j^{-1} \hat{\delta}_j$ has the usual asymptotic χ_S^2 null law. To derive the degrees of freedom, factor $V_j = L_j L_j^\top$. Under the null, $L_j^{-1} \hat{\delta}_j$ is asymptotically a vector of S independent standard normals, and Q_j is its squared norm. For one contrast, preserve the paper's signed statistic:

$$W_j = \frac{(C \otimes B_j^\top) \hat{\beta} - \delta_{0j}}{\sqrt{(C \otimes B_j^\top) V_\beta (C \otimes B_j^\top)^\top}}, \quad Q_j = W_j^2.$$

The manuscript's (2.15)–(2.16) use $\delta_{0j} = 0$. The signed W_j is asymptotically $N(0, 1)$, whereas its square is asymptotically χ_1^2 ; these names are not interchanged here. Reduce redundant constraints explicitly rather than using a pseudoinverse to hide an unidentified test. Contrast rows need not sum to one: differences have row sum zero.

11.2 The vectorized Wald map uses covariance diagonals

For $S = 1$ and a zero null, set $T = C \otimes B \in \mathbb{R}^{N \times RP}$. Then $\hat{\eta}_C = T \hat{\beta}$ and its covariance estimate is $TV_\beta T^\top$, an $N \times N$ matrix. Only its diagonal contains the marginal variances required for the paper's vector of signed statistics:

$$\mathbf{W} = \hat{\eta}_C / \sqrt{\text{diag}(TV_\beta T^\top)}.$$

Here diag extracts the diagonal of a square matrix, and division and square root are elementwise. The full covariance matrix under a square root is not an N -vector of standard errors; this makes that step in (2.16) dimensionally explicit. For a two-sided normal reference, $P_j = 2\{1 - \Phi(|W_j|)\}$, where Φ is the standard normal distribution function. The diagonal can be computed by the selected solves above without forming the full map covariance.

11.3 Comparing independently fitted groups

The manuscript’s group comparison can fit the SV model independently in disjoint groups with coefficient vectors β_g, β_h . If their data and fitting decisions are genuinely independent, their joint covariance is $\text{blkdiag}(V_g, V_h)$. For a fixed covariate-effect contrast $c \in \mathbb{R}^R$,

$$T_j = [c^\top \otimes B_j^\top \quad -c^\top \otimes B_j^\top], \quad V_{\delta j} = (c^\top \otimes B_j^\top)(V_g + V_h)(c \otimes B_j).$$

Equivalently, stack $(\beta_g^\top, \beta_h^\top)^\top$ and use $C_{\text{pair}} = (c^\top, -c^\top) \in \mathbb{R}^{1 \times 2R}$. This is a row contrast; a terminal transpose would instead make it a $2R \times 1$ column. Unequal group sizes enter V_g, V_h ; equality of a scientific effect under a null is not a claim that finite-sample estimates or their variances are independent of sample size. If groups share subjects, fitted nuisance coefficients, or a coupled selection procedure, zero cross-covariance needs reconsideration. This separately fitted SV comparison is not the shared-global multi-group CBMR parameterization of the other paper.

11.4 Polynomial covariates and nonlinear intensity summaries

If an age covariate t enters through t, t^2, t^3 , label those three components of the voxel coefficient vector as $\varphi_{j1}, \varphi_{j2}, \varphi_{j3}$. Their log effect is $\varphi_{j1}t + \varphi_{j2}t^2 + \varphi_{j3}t^3$ and its local derivative is

$$\frac{\partial \eta_j(t)}{\partial t} = \varphi_{j1} + 2t\varphi_{j2} + 3t^2\varphi_{j3}.$$

This is a linear coefficient contrast at any specified t , with covariate row $(1, 2t, 3t^2)$ in the polynomial coordinates. A finite change from t_0 to t_1 instead uses $(t_1 - t_0, t_1^2 - t_0^2, t_1^3 - t_0^3)$. Its intensity ratio is the exponential of that log contrast; it is not generally the exponential of a single linear-age coefficient.

For a specified subject design row $Z_*^\top$, put $x_{*j} = Z_* \otimes B_j$. Then $\mu_{*j} = e^{x_{*j}^\top \beta}$ has Jacobian $\partial \mu_{*j} / \partial \beta^\top = \mu_{*j} x_{*j}^\top$. Therefore $\text{Cov}(\hat{\mu}_{*j}, \hat{\mu}_{*k}) \approx \mu_{*j} \mu_{*k} x_{*j}^\top V_\beta x_{*k}$. For a scalar log contrast δ , the delta-method variance of $e^{\hat{\delta}}$ is approximately $e^{2\delta} \text{Var}(\hat{\delta})$. These Jacobians are exact; the covariance transformations are first-order approximations. An intensity ratio is a risk ratio only when the fitted intensities can validly be interpreted as the conditional binary means. For the explicitly conditioned logistic model, $e^{\hat{\delta}}$ is instead an odds ratio.

11.5 Multiplicity and bootstrap remain separate statistical steps

Pointwise Wald p -values do not automatically control error over many voxels. Benjamini–Hochberg targets FDR under its required dependence conditions, not general FWER control. Neither faster Hessians nor a different spatial basis proves those conditions or calibrates a finite-sample tail.

A nonparametric bootstrap should resample independent subjects with their whole binary images and covariates, or the appropriate larger independent clusters. A fixed-design parametric bootstrap keeps covariates fixed and simulates the specified response model. For a null test, fit and simulate the null rather than the unrestricted alternative, and repeat the estimation, dispersion fitting, and data-driven tuning steps that define the reported estimator. If Poisson/NB counts are simulated, the experiment assesses that count-model bootstrap; it does not silently become a Bernoulli lesion bootstrap. A spatial binary simulator must preserve the intended within-subject dependence. Failed refits must be reported and diagnosed, not silently removed from the reference distribution.

12 Supplement S2.3: a complete toy slope-map variance derivation

12.1 Translate the supplement's local symbols and coefficient order

Only in this section, follow S2.3 with *scalar* subject values Z_i and spatial values B_j , and two-component vectors

$$x_i = (1, Z_i)^\top, \quad b_j = (1, B_j)^\top, \quad \beta_{\text{toy}} = (\alpha, \gamma, \delta, \theta)^\top.$$

These scalar Z_i, B_j are not the main text's design columns. Likewise, the supplement's α is now a log-intercept, not NB dispersion; γ, δ, θ are scalar toy coefficients, not the initialization vector or log-dispersion coordinate. We retain these local names so each step can be compared with S2.3. Let $\mathbf{X}$ and $\mathbf{B}$ have rows $x_i^\top$ and $b_j^\top$; these are the local matrices called X and B in that supplement.

Expand the supplement's basis-first Kronecker product:

$$b_j \otimes x_i = (1, Z_i, B_j, Z_i B_j)^\top, \quad \log \mu_{ij} = \alpha + \gamma Z_i + \delta B_j + \theta Z_i B_j.$$

The main text instead orders $x_i \otimes b_j = (1, B_j, Z_i, Z_i B_j)^\top$. Thus its map-major coefficient vector is $(\alpha, \delta, \gamma, \theta)^\top$, obtained by interchanging coordinates two and three. If Π is that permutation matrix, then $\beta_{\text{main}} = \Pi \beta_{\text{toy}}$ and $I_{\text{main}} = \Pi I_{\text{toy}} \Pi^\top$. Contrasts must be permuted with the coefficients; the variance of the same scientific slope is then unchanged.

12.2 Separate the subject and spatial Gram matrices

Assume $\sum_i Z_i = 0$, as in the supplement, and define

$$S_Z = \sum_i Z_i^2, \quad S_B = \sum_j B_j^2.$$

Direct multiplication, without mixing the two sets of indices, gives

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} M & 0 \\ 0 & S_Z \end{pmatrix}, \quad \mathbf{B}^\top \mathbf{B} = \begin{pmatrix} N & \sum_j B_j \\ \sum_j B_j & S_B \end{pmatrix}.$$

Subject centering eliminates $\sum_i Z_i$ from the first matrix. It does not replace that zero by a spatial sum $\sum_j B_j$, and does not replace the subject count M by the voxel count N .

The exact Poisson information in the supplement's order is

$$I_{\text{toy}} = \sum_{i,j} \mu_{ij} (b_j b_j^\top) \otimes (x_i x_i^\top).$$

If the subject-dependent log effects $(\gamma + \theta B_j) Z_i$ are small, approximate $\mu_{ij} \approx \mu_j = \exp(\alpha + \delta B_j)$. This equality is exact at $\gamma = \theta = 0$. Pulling the separate sums out of the Kronecker product gives

$$I_{\text{toy}} \approx \left(\sum_j \mu_j b_j b_j^\top \right) \otimes \left(\sum_i x_i x_i^\top \right).$$

Write the supplement's scalar moments A, B, C as

$$A_\mu = \sum_j \mu_j, \quad B_\mu = \sum_j B_j \mu_j, \quad C_\mu = \sum_j B_j^2 \mu_j,$$

to avoid collisions with the spatial design and contrast matrix. With $K_\mu = \begin{pmatrix} A_\mu & B_\mu \\ B_\mu & C_\mu \end{pmatrix}$ and $D_Z = \text{diag}(M, S_Z)$, the four-by-four information is

$$I_{\text{toy}} \approx K_\mu \otimes D_Z = \begin{pmatrix} MA_\mu & 0 & MB_\mu & 0 \\ 0 & S_Z A_\mu & 0 & S_Z B_\mu \\ MB_\mu & 0 & MC_\mu & 0 \\ 0 & S_Z B_\mu & 0 & S_Z C_\mu \end{pmatrix}. \quad (12.1)$$

12.3 Invert the two factors and project to the slope

The slope at voxel j is the supplement's $m_j = \gamma + B_j \theta = (b_j \otimes c)^\top \beta_{\text{toy}}$, where $c = (0, 1)^\top$. Assume $S_Z > 0$ and $\Delta_\mu = A_\mu C_\mu - B_\mu^2 > 0$. Then

$$K_\mu^{-1} = \frac{1}{\Delta_\mu} \begin{pmatrix} C_\mu & -B_\mu \\ -B_\mu & A_\mu \end{pmatrix}, \quad D_Z^{-1} = \text{diag}(1/M, 1/S_Z).$$

The mixed-product identity gives

$$\text{Var}(\hat{m}_j) \approx (b_j \otimes c)^\top (K_\mu^{-1} \otimes D_Z^{-1}) (b_j \otimes c) \quad (12.2)$$

$$= (b_j^\top K_\mu^{-1} b_j) (c^\top D_Z^{-1} c) \quad (12.3)$$

$$= \frac{C_\mu - 2B_\mu B_j + A_\mu B_j^2}{S_Z(A_\mu C_\mu - B_\mu^2)}. \quad (12.4)$$

The approximation combines the small-slope information factorization with inverse-information asymptotics for the estimator. The matrix inversion and contrast projection are exact algebra.

For a more interpretable form, define

$$\bar{B}_\mu = B_\mu / A_\mu, \quad S_{BB,\mu} = \sum_j \mu_j (B_j - \bar{B}_\mu)^2 = C_\mu - B_\mu^2 / A_\mu.$$

Expanding and completing the square in the numerator yields

$$\text{Var}(\hat{m}_j) \approx \frac{1}{S_Z} \left\{ \frac{1}{A_\mu} + \frac{(B_j - \bar{B}_\mu)^2}{S_{BB,\mu}} \right\}.$$

The determinant condition is $A_\mu S_{BB,\mu} > 0$: the spatial feature must vary on positive-weight voxels. Neither a constant spatial feature nor a constant subject covariate can be repaired by an inverse formula.

12.4 Constant means, pooling, and a correction to the printed formula

When $\mu_j \approx \bar{\mu}$, let $\bar{B} = N^{-1} \sum_j B_j$ and $S_{BB} = \sum_j (B_j - \bar{B})^2$. Then $A_\mu = N\bar{\mu}$, $\bar{B}_\mu = \bar{B}$, and $S_{BB,\mu} = \bar{\mu} S_{BB}$, giving

$$\text{Var}(\hat{m}_j) \approx \frac{1}{\bar{\mu} S_Z} \left\{ \frac{1}{N} + \frac{(B_j - \bar{B})^2}{S_{BB}} \right\}. \quad (12.5)$$

The extra factor involving $NS_Z - (\sum_j B_j)^2$ in the supplied S2.3 derivation does not follow from its centered subject design. It mixes a voxel moment into the subject Gram matrix. In particular the final denominator must not replace S_Z by $S_Z - N\bar{B}^2$. Equations (12.1)–(12.5) show where the two dimensions remain separate.

For a hand check, take $M = 3$, $Z_i = (-1, 0, 1)$, $N = 4$, $B_j = (0, 1, 2, 3)$, and $\bar{\mu} = 0.1$. Then $S_Z = 2$, $\bar{B} = 1.5$, and $S_{BB} = 5$. The corrected inverse-information slope variances are $(3.5, 1.5, 1.5, 3.5)$; the erroneous factor $S_Z - N\bar{B}^2 = -7$ would give negative values. These are algebraic audit values, not a claim of accurate small-sample inference with only three subjects.

Under the same independent Poisson assumptions, separate voxelwise intercept/slope fits have inverse-information slope variance $1/(\bar{\mu}S_Z)$. The spatial result multiplies it by $h_j = 1/N + (B_j - \bar{B})^2/S_{BB}$. These are the diagonal leverages of the two-column spatial design: $0 \leq h_j \leq 1$ and $\sum_j h_j = 2$. This quantifies pooling when the spatial slope restriction is appropriate; it is not a guarantee under a misspecified slope map or residual spatial dependence. For binary or correlated lesion images, the applicable score-variability calculation remains the subject sandwich, not an automatic identification of Poisson information with variance.

13 The mass-univariate logistic and Firth comparator

13.1 Logistic score and Hessian at one voxel

For the comparator in manuscript S1.1, fit each voxel separately. Its symbols $(X, x_i, \beta_j, \eta_{ij})$ correspond here to $(Z, Z_i, \varphi_j, \zeta_{ij})$, respectively. This preserves the main text's design and voxel-coefficient names while reserving X, β for the full spatial model. The comparator's φ_j is fitted with a logistic link here; it is not asserted to equal the Poisson MUM estimate in (2.5). Use coefficient $\varphi_j \in \mathbb{R}^R$, predictor $\zeta_{ij} = Z_i^\top \varphi_j$, and $p_{ij} = e^{\zeta_{ij}}/(1 + e^{\zeta_{ij}})$. The Bernoulli log-likelihood is

$$\ell_j = \sum_i \{Y_{ij} \log p_{ij} + (1 - Y_{ij}) \log(1 - p_{ij})\} \quad (13.1)$$

$$= \sum_i \{Y_{ij} \zeta_{ij} - \log(1 + e^{\zeta_{ij}})\}. \quad (13.2)$$

Since $\partial_\zeta p = p(1 - p)$,

$$U_j = Z^\top (Y_{\cdot j} - p_{\cdot j}), \quad H_j = I_j = Z^\top \text{diag}\{p_{\cdot j}(1 - p_{\cdot j})\}Z.$$

This $R \times R$ matrix is for one voxel's comparator, not the $RP \times RP$ spatial model. Restricting the logistic coefficient field to a basis would instead use the full spatial design and weights $p_{ij}(1 - p_{ij})$. Neither formulation has the Poisson weight μ_{ij} .

13.2 Firth's determinant term is not a fixed quadratic penalty

The manuscript's Firth comparator maximizes $\ell_{F,j} = \ell_j + \frac{1}{2} \log \det I_j$. Assume I_j is nonsingular. The identity $d \log \det I = \text{tr}(I^{-1} dI)$ gives

$$(U_{F,j})_a = (U_j)_a + \frac{1}{2} \text{tr}(I_j^{-1} I_{j,a}), \quad I_{j,a} = \partial_{\varphi_{j,a}} I_j. \quad (13.3)$$

Write $w_i = p_i(1 - p_i)$, suppressing the fixed voxel j . Then

$$\partial_{\zeta_i} w_i = w_i(1 - 2p_i), \quad (13.4)$$

$$\partial_{\zeta_i}^2 w_i = w_i(1 - 2p_i)^2 - 2w_i^2, \quad (13.5)$$

$$I_{j,a} = \sum_i w_i(1 - 2p_i) Z_{ia} Z_i Z_i^\top, \quad (13.6)$$

$$I_{j,ab} = \sum_i \{w_i(1 - 2p_i)^2 - 2w_i^2\} Z_{ia} Z_{ib} Z_i Z_i^\top. \quad (13.7)$$

The first derivative also has a directly interpretable leverage form. Define $h_i = w_i Z_i^\top I_j^{-1} Z_i$. Using $\text{tr}(I_j^{-1} Z_i Z_i^\top) = Z_i^\top I_j^{-1} Z_i$,

$$\frac{1}{2} \text{tr}(I_j^{-1} I_{j,a}) = \frac{1}{2} \sum_i w_i(1 - 2p_i) Z_{ia} Z_i^\top I_j^{-1} Z_i \quad (13.8)$$

$$= \sum_i h_i(1/2 - p_i) Z_{ia}, \quad (13.9)$$

$$U_{F,j} = Z^\top \{Y_{\cdot j} - p_{\cdot j} + h \odot (1/2 - p_{\cdot j})\}. \quad (13.10)$$

Here $Y_{\cdot j} = (Y_{1j}, \dots, Y_{Mj})^\top$ is the vector at voxel j , not the initialization's summed margin $(Y_B)_j$. For a full-rank design, $\sum_i h_i = \text{tr}(I_j^{-1} I_j) = R$. This expands the determinant adjustment, rather than treating it as a constant ridge term. Differentiating $I^{-1} I = I_R$ gives $\partial_b I^{-1} = -I^{-1} I_b I^{-1}$. Therefore the full observed information of the penalized comparator is

$$(H_{F,j})_{ab} = (H_j)_{ab} - \frac{1}{2} \text{tr}\{I_j^{-1} I_{j,ab} - I_j^{-1} I_{j,b} I_j^{-1} I_{j,a}\}. \quad (13.11)$$

The determinant term depends on the coefficients and has its own curvature. It is not the fixed roughness penalty from the published coordinate-based CBMR model. The sampling covariance of a bias-reduced estimator also requires its inferential assumptions; an inverse penalized Hessian is not a universal finite-sample standard-error formula.

14 Numerical conditioning and explicitly optional regularization

14.1 Symmetric eigenvalues replace an unnecessary general SVD

For symmetric nonsingular $H = Q\Lambda Q^\top$, $H^\top H = Q\Lambda^2 Q^\top$, so its singular values are $|\lambda_a|$. Thus

$$\kappa_2(H) = \frac{\max_a |\lambda_a|}{\min_a |\lambda_a|}.$$

For SPD information the absolute values are unnecessary. An indefinite matrix may have a finite condition number but not a valid inverse-covariance interpretation. For disconnected blocks take the largest absolute eigenvalue over all blocks divided by the smallest over all blocks, not the largest within-block condition number. For example, blocks I and $10^6 I$ each have condition number one, while their combined matrix has condition number 10^6 .

For exact $P_0 = K_Z \otimes K_B$, $\kappa_2(P_0) = \kappa_2(K_Z)\kappa_2(K_B)$. This diagnostic describes that preconditioner; it need not describe the final nonseparable information. Solving with a factorization is preferable to explicitly forming an inverse when only directions or contrasts are requested. Symmetrizing roundoff does not repair genuine singularity.

14.2 A fixed quadratic penalty would be an additional model choice

The manuscript's core spatial model reduces dimension using a fixed basis; it does not thereby acquire the other paper's roughness penalty. If one deliberately adds $\mathcal{P}(\beta) = \frac{1}{2}\beta^\top K\beta$ with fixed symmetric $K \succeq 0$, then

$$\nabla(-\ell + \mathcal{P}) = -U + K\beta, \quad H_{\text{pen}} = H + K.$$

Linearizing $U(\hat{\beta}) - K\hat{\beta} = 0$ gives

$$(H + K)(\hat{\beta} - \beta_0) \approx U(\beta_0) - K\beta_0.$$

Its local sampling covariance is $(H + K)^{-1} J (H + K)^{-\top}$, not automatically $(H + K)^{-1}$, and its local bias includes $-(I + K)^{-1} K\beta_0$. Under a correctly specified count likelihood,

$$(I + K)^{-1} - (I + K)^{-1} I (I + K)^{-1} = (I + K)^{-1} K (I + K)^{-1}.$$

Inverse penalized curvature is instead the usual local Gaussian posterior curvature when the penalty is interpreted as a prior. Neither expression includes data-selected penalty uncertainty.

Adding a numerical ridge only when computing a covariance changes the inverse without changing the estimator's fitted equation. It cannot be described as new data identification. For PSD H, K , $H + K \succ 0$ exactly when their null spaces share no nonzero direction; a roughness penalty may leave a constant confounding direction untouched.

14.3 Computational accounting for the manuscript's alternatives

The following orders use dense arithmetic and a fixed basis. They separate fitting calculations from the covariance output; they are not measured timings.

Calculation	Leading operation count and storage implication
Exact batched Poisson mean and score	$O(MNP + MRP)$ work with subject batching; no explicit $MN \times RP$ design is needed
Exact independent-cell information	$O(MNP^2 + MR^2P^2)$ for studywise basis products and block assembly; dense output still has $(RP)^2$ entries
Fixed separable Poisson preconditioner	Factor work $O(P^3 + R^3)$ after forming its two weighted Gram matrices; factor storage $O(P^2 + R^2)$
Poisson Taylor score after fixed projections	$O(NP^2 + R^2P + RP^2)$ per iterate for K_a , matrix products and the new expansion point; no full current $M \times N$ mean array
Dense final information factorization	$O((RP)^3)$ work and $O((RP)^2)$ storage unless additional exact structure is available
k contrast covariance solves	$O(k(RP)^2)$ after a dense factorization, with $RP \times k$ solve output instead of a full inverse
Exact Hessian–vector product	$O(MRP + MNP)$ per product with current weights; no dense information matrix, but iterative solves need convergence control

The exact-score order follows by evaluating $\tilde{\eta} = (Z\tilde{\beta})B^\top$ and $Z^\top(\tilde{Y} - \tilde{\mu})B$, choosing that multiplication order. The information order follows from one $O(NP^2)$ weighted basis product per subject and its distribution into R^2 coefficient blocks. For the Taylor row, cache $Z^\top\tilde{Y}B$, $Z^\top\tilde{Z}$, and $Z^\top\mathbf{1}_M$ once. At an iterate form a and $K_a = B^\top \text{diag}(a)B$, then multiply $(Z^\top\tilde{Z})\tilde{\beta}K_a$ using matrices of sizes R and P . These costs do not apply unchanged to the data-dependent NB correction in (8.11). Input storage, sparse count projections, basis sparsity, and batching affect actual resource usage, and a small approximate score cost does not establish its accuracy.

15 Hand calculations and a paper-specific review checklist

15.1 A two-subject, two-voxel spatial model

Take $B = I_2$ and $Z = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$. Order $\beta = (\beta_{1,1}, \beta_{1,2}, \beta_{2,1}, \beta_{2,2})^\top$, with the first covariate the intercept. At $\beta = 0$, all $\mu_{ij} = 1$ and

$$X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \quad H_P = \begin{pmatrix} 2 & 0 & 1 & 0 \\ 0 & 2 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}.$$

This is an algebraic evaluation point, not a recommended fitted binary mean. With binary counts $(0, 1, 1, 0)$ and $\alpha = 1$, the NB observed weights are $(1, 2, 2, 1)/4$, yielding

$$H_{\text{NB}} = \begin{pmatrix} 3/4 & 0 & 1/2 & 0 \\ 0 & 3/4 & 0 & 1/4 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/4 & 0 & 1/4 \end{pmatrix}, \quad I_{\text{NB}} = \frac{1}{2}H_P.$$

Each entry is the sum of four weighted design-row products. This makes the observed/expected distinction directly checkable without a numerical differentiation package.

15.2 Decisions that must be explicit when reading the manuscript

Location	Required interpretation or correction
S1.2 versus (2.8)	Conditioning Poisson on zero/one gives logistic; the unconditioned Poisson objective is different
(2.9)–(2.10)	NB is a count working model on binary observations, not a literal positive-dispersion Bernoulli family
(2.11), Algorithm 1	Identify the additive intercepts and embed the P -vector fit into PR coordinates explicitly
(2.12), Algorithm 1	NB score has the denominator $1 + \alpha\mu$
(2.13)	Exact for separable Poisson weights; approximate for general NB
Fixed-step iteration	SPD preconditioning alone does not guarantee convergence of an undamped unit-step update
(2.14), S2.1	Use centered $Q_M Z$, unweighted B in the data projection, vector β in the Kronecker term, and preserve the single-weight $B^\top \tilde{B}$ product
(2.14), moving expansion point	Its approximate score can have a nonsymmetric Jacobian; it is not automatically a likelihood gradient
(2.15)–(2.16)	Contrasts need not sum to one; ranks and dimensions determine degrees of freedom. Keep W_j signed and use marginal covariance diagonals for the vectorized standard errors
(2.18)	Use the family’s own subject score and matching bread
S2.2	QR’s $\mathcal{R}$ is upper triangular; retain the column-stacked scores and correct two-solve order
S2.3	Translate the basis-first coefficient order, keep the subject Gram factor $\text{diag}(M, S_Z)$, and do not insert voxel moments into it

These qualifications distinguish mathematical changes from performance improvements. They do not claim to assess every simulation, empirical conclusion, or implementation associated with the manuscript. The count-model symbolic identities in `proofs/appendix.py` and related repository modules support parts of this exposition; the new paper-specific text and corrections are not automatically covered by the existing Lean inventory.

15.3 Editable source and independent build

This document is self-contained: it does not require the combined appendix or the other paper’s appendix to be read. Its editable source is `docs/cbmr_paper_appendix.tex`. From the repository root:

```
make cbmr-paper-appendix
```

Build both paper-specific appendices with `make paper-appendices`. The original papers and combined expanded appendix are preserved.